themselves to radiation doses of many Sv a year. It is reported – though supressed at that time – that the health of many of these young people rapidly deteriorated so much that they had to leave, some having accured anemia and other radiation related deceases, while others worked for decades without any obvious detrimental health effects. The explanation for this difference is assumed to be due to differences in the individuals immune defence system.

A similar conclusion can be drawn from Figure 15.6.A, where the number of cell abberations – measured a long time after exposure – varies with at least a factor of 5 (i.e. 500% !) for the same dose.

15.12. THE DOSE-EFFECT CURVE

Figure 15.11 shows a number of hypothetical dose-effect relations: The "unmeasurable range" is indicated within the circle. The dashed-dotted line outside the circle indicates the uncertainty in the "measurable range". Line a is based on the by ICRP recommended NTLE (No Threshold Linear Effect) hypothesis and the message is clear: the risk is zero only at zero radiation dose. Curve b indicates a threshold around 50 mSv,

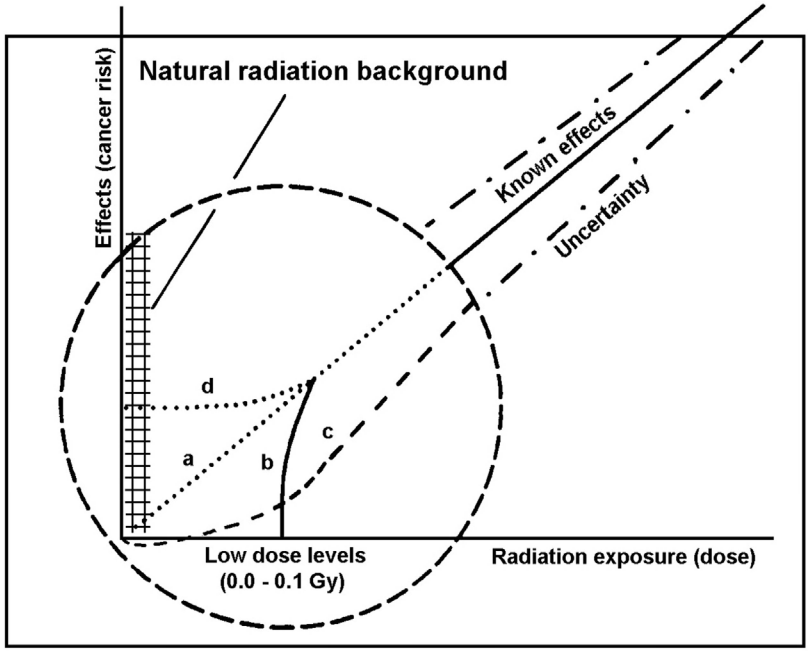


Figure 15.11 Dose effect relations in the known (with spread) and in the unknown (within circle) range, and some hypotheses.

below which there is no increase in cancer (or other radiation induced diseases); many radiologists support this hypothesis. Curve d assumes that there is a constant risk at the lowest doses. Curve c illustrates the "quadratic-linear" model, which presently seems to be favored by several radiation protection agencies (incl. ICRP), who assume that the slope near zero is one half of the slope at higher doses and dose rates. As this slope is unknown, it could as well be less. It is obvious that the NTLE hypothesis is wrong av very high doses because the effects saturate at some high dose level.

Many studies suggest the existence of a dose threshold level around 100 mSv, below which no health effects from radiation are observed. Other studies claim that health effects are observed even in the range 10–50 mSv. This has lead to "schools" of radiation health experts, those who claim that the NTLE thypothesis is scientifically validated, and those who claim that it is not. As an example we may study the effect of radiation on mice as shown by Ullrich and Storer 1979. It is clear from their study that a small radiation dose is not only harmless but actually beneficial for the health of the mice, see Figure 15.12. In conclusion, due to uncertainties the different interpretations of the low dose effects may be drawn as in Figure 15.11.

Figures 15.6, 15.8, and 15.9 show that it is impossible to assess the risk to humans at the 1–5 mSv level, which is the range of the natural radiation background. As epidemiological investigations seem to fall short, it has been suggested (Sondhaus, Yalow, Sagan, Cohen etc) that the only way to resolve this question is through *in vivo* radiation studies at the molecular level.

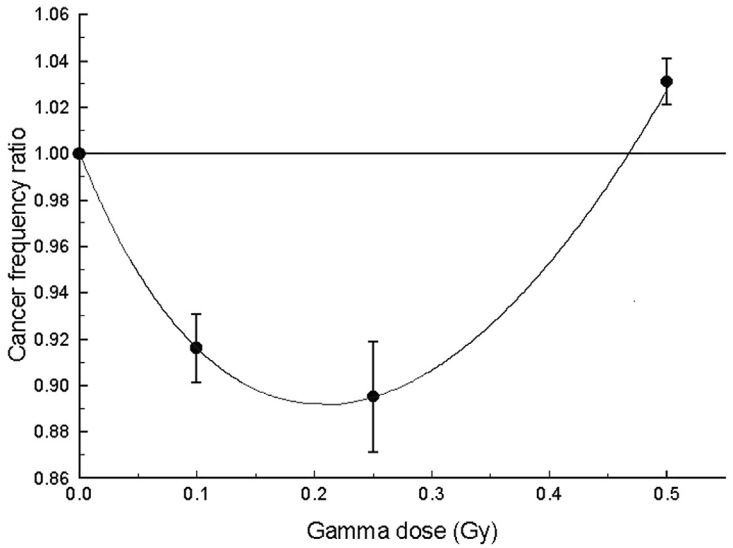


Figure 15.12 The effect of low doses of gamma radiation on the frequency of all cancers in mice, irradiated/unirradiated *(After Ullrich and Storer, 1979).*

15.13. REGULATORY RECOMMENDATIONS AND PROTECTION STANDARDS

15.13.1. Risk of cancer induction by radiation

The International Commission on Radiation Protection (ICRP) was formed in 1928 and has been the main international organization solely devoted to recommendations to prevent humans from being harmed by ionizing radiation. Their recommendations form the basis for national regulatory decisions.

In 1990, the ICRP concluded (Publ. # 60) that *the risk of cancer induction, L_c, is 5% per Sv* or

$$L_c = 0.05\ H_E \tag{15.5}$$

for low-levels of low-LET radiation (i.e. NTLE is assumed). Thus, for a dose of 20 Sv, the probability of acquiring cancer is set at 100%. Here L_c is the probability for an individual to acquire (and die) of cancer if exposed to the effective equivalent dose H_E (§15.4). This rule is based on the probabilities in Table 15.10.

In §15.6.1, it was stated that the lethal dose for instantaneous low-LET radiation is 10 Sv. The value 5%/Sv is half as large, which is a concession to the fact that biological harm depends on dose-rates as described in §15.5. In ICRP terminology, the *dose reduction factor* is chosen to be 2, although UNSCEAR 1993 (p.688) notes that it is more likely 2−10.

UNSCEAR collaborates closely with the United Nations Environment Program (UNEP), the World Health Organization (WHO), the International Atomic Energy Agency (IAEA), the International Commission on Radiation Units and Measurements (ICRU) and the ICRP.

In the light of previous discussion, one may state that the ICRP rules are "extra-safe". However, it is unlikely that the risk is underestimated, which is an important safety aspect. The ICRP does not recognize a threshold value, below which there is no harm at all. The ICRP notes (see e.g. BEIR V or UNSCEAR 1993) that the straight line a in Figure 15.11 may overestimate the true risks at low radiation doses and low dose rates, but should nevertheless be adhered to for safety reasons. However, it should not be used for prediction of cancer induction in large populations.

Table 15.10 Nominal probability coefficients for stochastic effects in detriment percent per Sv (10^{-2} Sv^{-1}) after exposure to radiation at low dose rate (ICRP 2007)

Exposed population	Fatal cancer	Heritable effects	Total detriment
Adult	4.1	0.1	4.2
Whole population	5.5	0.2	5.7